torch.cholesky_inverse#

https://pytorch.org/docs/stable/generated/torch.cholesky_inverse.html

Description:

Computes the inverse of a complex Hermitian or real symmetric positive-definite matrix given its Cholesky decomposition. Let AAA be a complex Hermitian or real symmetric positive-definite matrix, and LLL its Cholesky decomposition such that: where LHL^{\text{H}}LH is the conjugate transpose when LLL is complex, and the transpose when LLL is real-valued. Computes the inverse matrix A−1A^{-1}A−1. Supports input of float, double, cfloat and cdouble dtypes. Also supports batches of matrices, and if AAA is a batch of matrices then the output has the same batch dimensions.

Function Signature

```
torch.cholesky_inverse(L, upper=False, *, out=None) → Tensor#

Parameters

Keyword Arguments
```

Parameters

Keyword

out (Tensor, optional) – output tensor. Ignored if None. Default: None.

Code Examples

```
>>> A = torch.randn(3, 3)
>>> A = A @ A.T + torch.eye(3) * 1e-3 # Creates a symmetric
positive-definite matrix
>>> L = torch.linalg.cholesky(A) # Extract Cholesky
decomposition
>>> torch.cholesky_inverse(L)
tensor([[ 1.9314,  1.2251, -0.0889],
        [ 1.2251,  2.4439,  0.2122],
        [-0.0889,  0.2122,  0.1412]])
>>> A.inverse()
tensor([[ 1.9314,  1.2251, -0.0889],
        [ 1.2251,  2.4439,  0.2122],
        [-0.0889,  0.2122,  0.1412]])

>>> A = torch.randn(3, 2, 2, dtype=torch.complex64)
>>> A = A @ A.mH + torch.eye(2) * 1e-3 # Batch of Hermitian
positive-definite matrices
>>> L = torch.linalg.cholesky(A)
>>> torch.dist(torch.inverse(A), torch.cholesky_inverse(L))
tensor(5.6358e-7)
```

Detailed Documentation

Documentation

Computes the inverse of a complex Hermitian or real symmetric positive-definite matrix given its Cholesky decomposition.

Let AAA be a complex Hermitian or real symmetric positive-definite matrix, and LLL its Cholesky decomposition such that:

where LHL^{\text{H}}LH is the conjugate transpose when LLL is complex, and the transpose when LLL is real-valued.

Computes the inverse matrix A–1A^{-1}A–1.

Supports input of float, double, cfloat and cdouble dtypes. Also supports batches of matrices, and if AAA is a batch of matrices then the output has the same batch dimensions.

L (Tensor) – tensor of shape (*, n, n) where * is zero or more batch dimensions consisting of lower or upper triangular Cholesky decompositions of symmetric or Hermitian positive-definite matrices.

upper (bool, optional) – flag that indicates whether LLL is lower triangular or upper triangular. Default: False

out (Tensor, optional) – output tensor. Ignored if None. Default: None.